

Problem 4

Definition of Events:

- Let T be the event that the chosen passerby is a Tourist, with $\mathbb{P}(T) = \frac{2}{3}$.
- Let B be the event that the chosen passerby is a Berliner, with $\mathbb{P}(B) = \frac{1}{3}$.
- Let C_i be the event that the i -th answer is correct.
- Let $W_i = C_i^c$ be the event that the i -th answer is wrong.

From the problem description, we have the conditional probabilities:

$$\begin{aligned}\mathbb{P}(C_i | T) &= \frac{3}{4}, & \mathbb{P}(W_i | T) &= \frac{1}{4} \\ \mathbb{P}(C_i | B) &= 0, & \mathbb{P}(W_i | B) &= 1\end{aligned}$$

The answers to repeated questions are conditionally independent given the type of passerby.

Part a)

Since T and B partition the sample space, we apply the Law of Total Probability (Theorem 2.1.7):

$$\begin{aligned}\mathbb{P}(C_1) &= \mathbb{P}(C_1 | T)\mathbb{P}(T) + \mathbb{P}(C_1 | B)\mathbb{P}(B) \\ &= \left(\frac{3}{4}\right) \left(\frac{2}{3}\right) + (0) \left(\frac{1}{3}\right) \\ &= \frac{1}{2}\end{aligned}$$

The probability that the first answer is correct is $\frac{1}{2}$.

Part b)

We ask the same question n times and receive the same answer every time. Let E_n be the event that all n answers are identical, X_n the event that all n answers are correct, and Z_n the event that all n answers are incorrect, formally:

$$\begin{aligned}X_n &= C_1 \cap C_2 \cap \cdots \cap C_n \\ Z_n &= W_1 \cap W_2 \cap \cdots \cap W_n \\ E_n &= X_n \cup Z_n\end{aligned}$$

Since X_n and Z_n are disjoint, we have $E_n = X_n \cup Z_n$ and $\mathbb{P}(E_n) = \mathbb{P}(X_n) + \mathbb{P}(Z_n)$. Using the Law of Total Probability (LTP) and conditional independence (CI), we compute the probabilities of X_n and Z_n :

$$\begin{aligned}\mathbb{P}(X_n) &= \mathbb{P}(C_1 \cap \cdots \cap C_n) \\ &\stackrel{(LTP)}{=} \mathbb{P}(C_1 \cap \cdots \cap C_n | T)\mathbb{P}(T) + \mathbb{P}(C_1 \cap \cdots \cap C_n | B)\mathbb{P}(B) \\ &\stackrel{(CI)}{=} \left(\prod_{i=1}^n \mathbb{P}(C_i | T)\right) \mathbb{P}(T) + \left(\prod_{i=1}^n \mathbb{P}(C_i | B)\right) \mathbb{P}(B) \\ &= \left(\frac{3}{4}\right)^n \left(\frac{2}{3}\right) + 0^n \left(\frac{1}{3}\right) = \frac{2}{3} \left(\frac{3}{4}\right)^n\end{aligned}$$

$$\begin{aligned}
\mathbb{P}(Z_n) &= \mathbb{P}(W_1 \cap \dots \cap W_n) \\
&\stackrel{(LTP)}{=} \mathbb{P}(W_1 \cap \dots \cap W_n \mid T)\mathbb{P}(T) + \mathbb{P}(W_1 \cap \dots \cap W_n \mid B)\mathbb{P}(B) \\
&\stackrel{(CI)}{=} \left(\prod_{i=1}^n \mathbb{P}(W_i \mid T) \right) \mathbb{P}(T) + \left(\prod_{i=1}^n \mathbb{P}(W_i \mid B) \right) \mathbb{P}(B) \\
&= \left(\frac{1}{4} \right)^n \left(\frac{2}{3} \right) + 1^n \left(\frac{1}{3} \right) = \left(\frac{1}{3} \right) \left(2 \left(\frac{1}{4} \right)^n + 1 \right)
\end{aligned}$$

We want to find the conditional probability that the given answers were correct, given that they were all identical. By definition of conditional probability:

$$\begin{aligned}
\mathbb{P}(X_n \mid E_n) &= \frac{\mathbb{P}(X_n \cap E_n)}{\mathbb{P}(E_n)} \stackrel{(X_n \subseteq E_n)}{=} \frac{\mathbb{P}(X_n)}{\mathbb{P}(E_n)} \\
&= \frac{\frac{2}{3} \left(\frac{3}{4} \right)^n}{\frac{2}{3} \left(\frac{3}{4} \right)^n + \left(\frac{1}{3} \right) \left(2 \left(\frac{1}{4} \right)^n + 1 \right)}
\end{aligned}$$

Multiplying the numerator and denominator by $3 \cdot 4^n$ to simplify the expression:

$$\begin{aligned}
\mathbb{P}(X_n \mid E_n) &= \frac{2 \cdot 3^n}{2 \cdot 3^n + 2 \cdot 1^n + 1 \cdot 4^n} \\
&= \frac{2 \cdot 3^n}{2 \cdot 3^n + 4^n + 2}
\end{aligned}$$

Now we calculate this probability for $n = 2, 3, 4$:

- For $n = 2$: $\frac{2 \cdot 3^2}{2 \cdot 3^2 + 4^2 + 2} = \frac{18}{18 + 16 + 2} = \frac{18}{36} = \frac{1}{2}$
- For $n = 3$: $\frac{2 \cdot 3^3}{2 \cdot 3^3 + 4^3 + 2} = \frac{54}{54 + 64 + 2} = \frac{54}{120} = \frac{9}{20}$
- For $n = 4$: $\frac{2 \cdot 3^4}{2 \cdot 3^4 + 4^4 + 2} = \frac{162}{162 + 256 + 2} = \frac{162}{420} = \frac{27}{70}$

Part c)

Let F be the event that the passerby gives the same answer three times and a different answer the fourth time. Based on the definitions in Part b), this event can be expressed as:

$$F = (X_3 \cap W_4) \cup (Z_3 \cap C_4)$$

Since the events $(X_3 \cap W_4)$ and $(Z_3 \cap C_4)$ are disjoint, we apply the σ -additivity of probability measures to find $\mathbb{P}(F)$:

$$\mathbb{P}(F) = \mathbb{P}(X_3 \cap W_4) + \mathbb{P}(Z_3 \cap C_4)$$

Calculating the individual probabilities using conditional independence:

$$\begin{aligned}
\mathbb{P}(X_3 \cap W_4) &= \mathbb{P}(X_3 \cap W_4 \mid T)\mathbb{P}(T) + \mathbb{P}(X_3 \cap W_4 \mid B)\mathbb{P}(B) \\
&= \mathbb{P}(X_3 \mid T)\mathbb{P}(W_4 \mid T)\mathbb{P}(T) + \mathbb{P}(X_3 \mid B)\mathbb{P}(W_4 \mid B)\mathbb{P}(B) \\
&= \left(\frac{3}{4} \right)^3 \left(\frac{1}{4} \right) \left(\frac{2}{3} \right) + (0)^3 (1) \left(\frac{1}{3} \right) \\
&= \frac{27}{64} \cdot \frac{1}{4} \cdot \frac{2}{3} + 0 = \frac{54}{768} = \frac{9}{128}
\end{aligned}$$

$$\begin{aligned}
\mathbb{P}(Z_3 \cap C_4) &= \mathbb{P}(Z_3 \cap C_4 \mid T)\mathbb{P}(T) + \mathbb{P}(Z_3 \cap C_4 \mid B)\mathbb{P}(B) \\
&= \mathbb{P}(Z_3 \mid T)\mathbb{P}(C_4 \mid T)\mathbb{P}(T) + \mathbb{P}(Z_3 \mid B)\mathbb{P}(C_4 \mid B)\mathbb{P}(B) \\
&= \left(\frac{1}{4}\right)^3 \left(\frac{3}{4}\right) \left(\frac{2}{3}\right) + (1)^3(0) \left(\frac{1}{3}\right) \\
&= \frac{1}{64} \cdot \frac{3}{4} \cdot \frac{2}{3} + 0 = \frac{6}{768} = \frac{1}{128}
\end{aligned}$$

Therefore, the probability of the observed event F is:

$$\mathbb{P}(F) = \frac{9}{128} + \frac{1}{128} = \frac{10}{128}$$

We want to find the conditional probability that the first answer is correct, given F . By definition of conditional probability:

$$\mathbb{P}(C_1 \mid F) = \frac{\mathbb{P}(C_1 \cap F)}{\mathbb{P}(F)}$$

Simplifying the intersection in the numerator:

$$\begin{aligned}
C_1 \cap F &= C_1 \cap [(X_3 \cap W_4) \cup (Z_3 \cap C_4)] \\
&= (C_1 \cap X_3 \cap W_4) \cup (C_1 \cap Z_3 \cap C_4)
\end{aligned}$$

Since $X_3 = C_1 \cap C_2 \cap C_3$, it follows that $C_1 \cap X_3 = X_3$, and since $Z_3 = W_1 \cap W_2 \cap W_3 = C_1^c \cap C_2^c \cap C_3^c$, the intersection $C_1 \cap Z_3 = \emptyset$ (the first answer cannot be both correct and incorrect). Thus:

$$C_1 \cap F = (X_3 \cap W_4) \cup \emptyset = X_3 \cap W_4$$

We had calculated that the probability of $X_3 \cap W_4$ is $\frac{9}{128}$. Finally, substituting this back into the conditional probability formula:

$$\mathbb{P}(C_1 \mid F) = \frac{\mathbb{P}(C_1 \cap F)}{\mathbb{P}(F)} = \frac{\mathbb{P}(X_3 \cap W_4)}{\mathbb{P}(F)} = \frac{\frac{9}{128}}{\frac{10}{128}} = \frac{9}{10}$$

□

Part d)

Let's define the following events:

- A : Lisa is walking in the right direction.
- A^c : Lisa is walking in the wrong direction.
- Y : The passerby answers "Yes" to Lisa's question.
- N : The passerby answers "No" to Lisa's question.

We also have that the probability of Lisa walking in the right direction is:

$$\mathbb{P}(A) = \epsilon$$

Therefore, the probability of her walking in the wrong direction is:

$$\mathbb{P}(A^c) = 1 - \epsilon$$

First, we determine the conditional probabilities of receiving a **"Yes" answer**. We assume that the accuracy of the passerby's answer (C_i, W_i) is stochastically independent of Lisa's actual direction (A):

- If Lisa is walking in the right direction, a "Yes" indicates the passerby gave a correct answer (the first time the passerby is asked the question). By independence, we have:

$$\mathbb{P}(Y | A) = \mathbb{P}(C_1 | A) = \mathbb{P}(C_1) = \frac{1}{2}$$

- If Lisa is walking in the wrong direction, a "Yes" indicates the passerby gave an incorrect answer. By independence, we have:

$$\mathbb{P}(Y | A^c) = \mathbb{P}(W_1 | A^c) = \mathbb{P}(W_1) = \frac{1}{2}$$

Since A and A^c partition the sample space, we use the Law of Total Probability (Theorem 2.1.7) to find the unconditional probability of receiving a "Yes" answer:

$$\begin{aligned}\mathbb{P}(Y) &= \mathbb{P}(Y | A)\mathbb{P}(A) + \mathbb{P}(Y | A^c)\mathbb{P}(A^c) \\ &= \frac{1}{2}(\epsilon) + \frac{1}{2}(1 - \epsilon) \\ &= \frac{1}{2}\epsilon + \frac{1}{2} - \frac{1}{2}\epsilon = \frac{1}{2}\end{aligned}$$

Now, we apply Bayes' Formula to find the probability that Lisa is walking in the right direction given she received a "Yes":

$$\mathbb{P}(A | Y) = \frac{\mathbb{P}(Y | A)\mathbb{P}(A)}{\mathbb{P}(Y)} = \frac{\frac{1}{2} \cdot \epsilon}{\frac{1}{2}} = \epsilon$$

Second, we apply the same logic for a "No" answer:

$$\mathbb{P}(N | A) = \mathbb{P}(W_1 | A) = \mathbb{P}(W_1) = \frac{1}{2}$$

$$\mathbb{P}(N | A^c) = \mathbb{P}(C_1 | A^c) = \mathbb{P}(C_1) = \frac{1}{2}$$

Applying the Law of Total Probability yields

$$\begin{aligned}\mathbb{P}(N) &= \mathbb{P}(N | A)\mathbb{P}(A) + \mathbb{P}(N | A^c)\mathbb{P}(A^c) \\ &= \frac{1}{2}(\epsilon) + \frac{1}{2}(1 - \epsilon) = \frac{1}{2}\end{aligned}$$

Applying Bayes' Formula yields:

$$\mathbb{P}(A | N) = \frac{\mathbb{P}(N | A)\mathbb{P}(A)}{\mathbb{P}(N)} = \frac{\frac{1}{2} \cdot \epsilon}{\frac{1}{2}} = \epsilon$$

Therefore, regardless of whether the first answer is "Yes" or "No", Lisa's posterior probability of walking in the right direction remains ϵ .

□

Part e)

Receiving two identical answers means she either receives the sequence "Yes, Yes" (event JJ) or "No, No" (event NN). From Part b), we defined X_2 as the event that the first two answers are correct, and Z_2 as the event that the first two answers are incorrect. Let us calculate their exact probabilities using the formulas derived in Part b):

$$\begin{aligned}\mathbb{P}(X_2) &= \frac{2}{3} \left(\frac{3}{4} \right)^2 = \frac{2}{3} \cdot \frac{9}{16} = \frac{18}{48} = \frac{3}{8} \\ \mathbb{P}(Z_2) &= \frac{1}{3} \left(2 \left(\frac{1}{4} \right)^2 + 1 \right) = \frac{1}{3} \left(\frac{2}{16} + \frac{16}{16} \right) = \frac{1}{3} \cdot \frac{18}{16} = \frac{6}{16} = \frac{3}{8}\end{aligned}$$

Notice that $\mathbb{P}(X_2) = \mathbb{P}(Z_2) = \frac{3}{8}$.

First, we analyze the event JJ:

- If Lisa is walking in the right direction (A), two "Yes" answers mean both answers were correct. Thus, $\mathbb{P}(\text{JJ} \mid A) = \mathbb{P}(X_2) = \frac{3}{8}$.
- If Lisa is walking in the wrong direction (A^c), two "Yes" answers mean both answers were incorrect. Thus, $\mathbb{P}(\text{JJ} \mid A^c) = \mathbb{P}(Z_2) = \frac{3}{8}$.

Using the Law of Total Probability:

$$\begin{aligned}\mathbb{P}(\text{JJ}) &= \mathbb{P}(\text{JJ} \mid A)\mathbb{P}(A) + \mathbb{P}(\text{JJ} \mid A^c)\mathbb{P}(A^c) \\ &= \frac{3}{8}(\epsilon) + \frac{3}{8}(1 - \epsilon) = \frac{3}{8}\epsilon + \frac{3}{8} - \frac{3}{8}\epsilon = \frac{3}{8}\end{aligned}$$

Applying Bayes' Formula:

$$\mathbb{P}(A \mid \text{JJ}) = \frac{\mathbb{P}(\text{JJ} \mid A)\mathbb{P}(A)}{\mathbb{P}(\text{JJ})} = \frac{\frac{3}{8} \cdot \epsilon}{\frac{3}{8}} = \epsilon$$

Second, we analyze the event NN:

- If Lisa is walking in the right direction (A), two "No" answers mean both answers were incorrect. Thus, $\mathbb{P}(\text{NN} \mid A) = \mathbb{P}(Z_2) = \frac{3}{8}$.
- If Lisa is walking in the wrong direction (A^c), two "No" answers mean both answers were correct. Thus, $\mathbb{P}(\text{NN} \mid A^c) = \mathbb{P}(X_2) = \frac{3}{8}$.

Using the Law of Total Probability:

$$\begin{aligned}\mathbb{P}(\text{NN}) &= \mathbb{P}(\text{NN} \mid A)\mathbb{P}(A) + \mathbb{P}(\text{NN} \mid A^c)\mathbb{P}(A^c) \\ &= \frac{3}{8}(\epsilon) + \frac{3}{8}(1 - \epsilon) = \frac{3}{8}\end{aligned}$$

Applying Bayes' Formula:

$$\mathbb{P}(A \mid \text{NN}) = \frac{\mathbb{P}(\text{NN} \mid A)\mathbb{P}(A)}{\mathbb{P}(\text{NN})} = \frac{\frac{3}{8} \cdot \epsilon}{\frac{3}{8}} = \epsilon$$

Therefore, whether she receives the sequence "Yes, Yes" or "No, No", Lisa's posterior probability of walking in the right direction remains exactly ϵ .

□

Part f)

We want to calculate Lisa's posterior probability of walking in the right direction (event A) after receiving three identical answers, which means either three "Yes" answers (event JJJ) or three "No" answers (event NNN). We calculate the probabilities of the events that the three answer are correct (X_3) and incorrect (Z_3) using our derived formulas in part b):

$$\begin{aligned}\mathbb{P}(X_3) &= \frac{2}{3} \left(\frac{3}{4}\right)^3 = \frac{2}{3} \cdot \frac{27}{64} = \frac{18}{64} = \frac{9}{32} \\ \mathbb{P}(Z_3) &= \frac{1}{3} \left(2 \left(\frac{1}{4}\right)^3 + 1\right) = \frac{1}{3} \left(\frac{2}{64} + \frac{64}{64}\right) = \frac{1}{3} \cdot \frac{66}{64} = \frac{22}{64} = \frac{11}{32}\end{aligned}$$

In this case, $\mathbb{P}(X_3) \neq \mathbb{P}(Z_3)$.

First, we analyze the event JJJ:

- If Lisa is walking in the right direction (A), three "Yes" answers mean all three answers were correct. Thus, $\mathbb{P}(\text{JJJ} \mid A) = \mathbb{P}(X_3) = \frac{9}{32}$.
- If Lisa is walking in the wrong direction (A^c), three "Yes" answers mean all three answers were incorrect. Thus, $\mathbb{P}(\text{JJJ} \mid A^c) = \mathbb{P}(Z_3) = \frac{11}{32}$.

Using the Law of Total Probability, we find the unconditional probability of receiving three "Yes" answers:

$$\begin{aligned}\mathbb{P}(\text{JJJ}) &= \mathbb{P}(\text{JJJ} \mid A)\mathbb{P}(A) + \mathbb{P}(\text{JJJ} \mid A^c)\mathbb{P}(A^c) \\ &= \frac{9}{32}(\epsilon) + \frac{11}{32}(1 - \epsilon) \\ &= \frac{9\epsilon + 11 - 11\epsilon}{32} = \frac{11 - 2\epsilon}{32}\end{aligned}$$

Applying Bayes' Formula, the probability she is walking in the right direction given JJJ is:

$$\begin{aligned}\mathbb{P}(A \mid \text{JJJ}) &= \frac{\mathbb{P}(\text{JJJ} \mid A)\mathbb{P}(A)}{\mathbb{P}(\text{JJJ})} \\ &= \frac{\frac{9}{32} \cdot \epsilon}{\frac{11 - 2\epsilon}{32}} = \frac{9\epsilon}{11 - 2\epsilon}\end{aligned}$$

Second, we analyze the event NNN:

- If Lisa is walking in the right direction (A), three "No" answers mean all three answers were incorrect. Thus, $\mathbb{P}(\text{NNN} \mid A) = \mathbb{P}(Z_3) = \frac{11}{32}$.
- If Lisa is walking in the wrong direction (A^c), three "No" answers mean all three answers were correct. Thus, $\mathbb{P}(\text{NNN} \mid A^c) = \mathbb{P}(X_3) = \frac{9}{32}$.

Using the Law of Total Probability, we find the unconditional probability of receiving three "No" answers:

$$\begin{aligned}\mathbb{P}(\text{NNN}) &= \mathbb{P}(\text{NNN} \mid A)\mathbb{P}(A) + \mathbb{P}(\text{NNN} \mid A^c)\mathbb{P}(A^c) \\ &= \frac{11}{32}(\epsilon) + \frac{9}{32}(1 - \epsilon) \\ &= \frac{11\epsilon + 9 - 9\epsilon}{32} = \frac{9 + 2\epsilon}{32}\end{aligned}$$

Applying Bayes' Formula, the probability she is walking in the right direction given NNN is:

$$\begin{aligned}\mathbb{P}(A \mid \text{NNN}) &= \frac{\mathbb{P}(\text{NNN} \mid A)\mathbb{P}(A)}{\mathbb{P}(\text{NNN})} \\ &= \frac{\frac{11}{32} \cdot \epsilon}{\frac{9+2\epsilon}{32}} = \frac{11\epsilon}{9+2\epsilon}\end{aligned}$$

Both formulas match the target expressions exactly.

□